

Esercizio 1. (Martina Apicella) Let M and N be elementarily homogeneous structures of the same cardinality λ . Suppose that $M \models \exists x p(x) \Leftrightarrow N \models \exists x p(x)$ for every $p(x) \subseteq L$ such that $|x| < \lambda$. Prove that the two structures are isomorphic.

Esercizio 2. (Beatrice Calderaro) Let λ be a regular cardinal $\geq |L|$. Let $\langle M_i : i < \lambda \rangle$ be an elementary chain such that M_{i+1} realizes every $p(x) \subseteq L(M_i)$, where $|x| = 1$, that is finitely consistent in M_i . Prove that the union of the chain is λ -homogeneous.

Esercizio 3. (Francesco Dam) Let $\varphi(x) \in L$. Prove that the following are equivalent

1. $\varphi(x)$ is equivalent (in $\mathcal{U}$) to some $\psi(x) \in L_{\text{qf}}$
2. $\varphi(a) \leftrightarrow \varphi(fa)$ for every partial embedding $f : \mathcal{U} \rightarrow \mathcal{U}$ and $a \in (\text{dom } f)^x$.

Give a direct proof by saturation (not through Theorem 10.3).

Esercizio 4. (Filippo Giugie) Let T be a complete theory without finite models in a language that consists only of unary predicates. Prove that T has elimination of quantifiers. Is the claim true if T is not complete?

Esercizio 5. (Daniele Lanciotti) Let T be the theory of *discrete linear orders*, that is, T extends the theory of linear orders T_{lo} with the following two axioms

dis $\uparrow$. $\exists z [x < z \wedge \neg \exists y x < y < z]$

dis $\downarrow$. $\exists z [z < x \wedge \neg \exists y z < y < x]$.

Let Δ be the set of formulas that contains (all alphabetic variants of) the formulas

$$x <_n y := \exists^{\geq n} z (x < z < y)$$

and their negations, for every $n > 0$.

Prove that the theory of discrete linear orders has Δ -elimination of quantifiers. Prove that the structure $\mathbb{Q} \times \mathbb{Z}$ ordered with the lexicographic order is a saturated model of T .

Esercizio 6. (Mattia Lo Mele) Let T be a consistent theory without finite models. Suppose that all completions of T are of the form $T \cup S$ for some set S of quantifier-free sentences. Apply Corollary 10.12 to prove that if all completions of T have elimination of quantifiers, so does T . Show that this fails when the completions of T have arbitrary complexity.

Esercizio 7. (Andrea Morello) A countable structure M is Δ -set-homogeneous if for every finite Δ -morphism $k : M \rightarrow M$ there is an $h \in \text{Aut}(M)$ such that $h[\text{dom } k] = \text{im } k$. Prove that the theory of Δ -set-homogeneous models is Δ -model-complete. Hint: prove that every prime $\{\exists\}\Delta$ -type is maximal.

Esercizio 8. (Gabriele Maiolino) Work in a monster model $\mathcal{U}$. Suppose that every formula $\varphi(x) \in L$ is equivalent to a some quantifier-free formula $\psi(x) \in L(\mathcal{U})$. Prove that T is Δ -model complete for Δ the set of formulas of the form $\forall y \exists z \vartheta(x, y, z)$ with $\vartheta(x, y, z) \in L_{\text{qf}}$.

Esercizio 9. Use Propopsition 9.27 to give a third quick proof of Proposition 9.26.